

AI Critique — HW06 API Testing

Status: CONFIRMED — accepted through [confirm critique](#) on 2026-08-28T23:00:19+07:00.

Evidence matrix

Evidence ID	Pool/Stage	AI output or omission	Ground-truth evidence	Human correction	Why AI failed	Source path
CRIT-001	Pool A / Stages 1–2	The generated FSM included no-request wait rows and cases combining multiple invalid conditions	Human audit classified 12 of 79 cases INVALID , including non-observable waits and mixed-condition state cases	Removed no-op rows or converted them into executable POST checks; split cases so one invalid class/state condition is isolated	The transition-table expansion optimized combinatorial coverage without enforcing API executability and test isolation on every row	Pool-A_FR03_Password_Reset/stage2-audit-proposal.md
CRIT-002	Pool A / Stage 3	Domain generation collapsed the documented password special-character set into one successful representative and omitted leading-zero OTP serialization	The SRS lists discrete allowed values; Stage 3 records a separate “Why AI missed it” explanation for each gap	Added six successful-reset representatives plus one conditional leading-zero token case	The model treated explicit discrete values as interchangeable and emphasized length/type boundaries	Pool-A_FR03_Password_Reset/stage3-extension-proposal.md
CRIT-003	Pool B / Stage 3	Initial techniques omitted stale totals after cart mutation, nested item/discount injection, query/body pollution, duplicate JSON keys, and a benign apostrophe control	Stage 3 identified seven non-duplicate gaps with attributable explanations; the final run later proved client-controlled pricing across 12 isolated cases	Added one domain, one temporal decision-table, and five security cases	Static Boolean modeling missed temporal state; attack-oriented generation under-sampled parser differentials and positive false-positive controls	Pool-B_FR08_Checkout/stage3-extension-proposal.md ; Pool-B_FR08_Checkout/stage5-bug-report-proposal.md
CRIT-004	Pool C / Stage 3	Initial coverage omitted a multi-byte 255-character boundary, exponent numeric encoding, array-valued role claims, duplicate Authorization headers, and three field-	The human-confirmed baseline and gap ledger supported seven new non-duplicate cases	Added two domain and five security representatives while preserving unspecified contracts	The model conflated equivalent semantic values/encodings and assumed shared parsing or binding behavior across fields	Pool-C_FR15_Update_Product/stage3-extension-proposal.md

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		specific SQL strings				
CRIT-005	Pools A/B/C / Stage 4	Generated automation contained double-slash request paths in Pools A/C, a StudentId versus StudentID mismatch in Pool B, and a serialized regex defect	Diagnostic runs showed route failures/header problems/assertion exceptions; corrected runs produced real Newman and state evidence	Discarded diagnostic results, corrected collection generation/scripts, reset disposable fixtures, reran, and classified runner defects separately from SUT defects	Code generation was syntactically plausible but not reliable across Postman URL serialization, case-sensitive variables, and escaped script text	Pool-A_FR03_Password_Reset/stage4-postman-proposal.md; Pool-B_FR08_Checkout/stage4-postman-proposal.md; Pool-C_FR15_Update_Product/stage4-postman-proposal.md
CRIT-006	Pools A/B/C / Stages 4–5	Useful AI contribution: structured collections, evidence ledgers, and failure grouping	Corrected evidence recorded 36 Pool A assertion failures, two Pool B defect root causes, and 37 Pool C failures partitioned exactly once	Humans confirmed or rejected classifications, preserved evidence gaps, avoided duplicate Issue 52, and kept issue posting manual	Technique-specific prompts and traceable IDs made breadth, reconciliation, and repeatable triage effective	Each Pool's stage4-postman-proposal.md, stage5-bug-report-proposal.md, and local ai_audit_report.md
CRIT-007	Cross-pipeline / CI/CD	The initial CI proposal lacked configured runtime secret evidence and later contained a results-directory working-directory defect	Real Actions diagnostics exposed the missing bundle and startup-path problem; the retained run then produced 96 passed and 63 failed cases	Created the reviewed secret, corrected the absolute results path, preserved exit codes/artifacts, and documented the student's evidence substitution without claiming literal two-run compliance	Repository/environment assumptions were incomplete until tested in the actual CI runtime	reports/HW06_CICD_Report.md; root ai_audit_report.md rows POST-AI-003-POST-AI-013

Evidence gaps carried forward

- Pool Stage 5 reports still contain human-owned failure-detail screenshot placeholders; this critique does not claim those screenshots or issue posts exist.
- The CI/CD report records a student-approved substitution: no 159/159 run exists, and the real failed run contains 63 failed cases rather than exactly one. Literal assignment compliance remains unmet unless the instructor accepts the deviation.
- Existing **POST-AI-001-POST-AI-013** rows are marked **CONFIRMED**, but the root Cross-pipeline Human Review Notes table lacks attributable per-row review timestamps and commands. Their recorded decisions are preserved without invented notes.

Critique

AI was most useful when it converted the specifications into separate domain, state, decision-table, security, and schema artifacts with traceable IDs. That structure supported data-driven Postman collections and disciplined triage. The final evidence linked AI-assisted tests to three Pool A defect drafts, two Pool B root causes, and two new Pool C reports plus one known issue.

However, the first-pass output was not safe to accept unchanged. Pool A's generated FSM included no-request waits and cases combining two invalid conditions; human audit classified 12 cases INVALID and required removal or isolation. Its domain pass also collapsed six explicitly allowed password special characters into one representative. Stage 3 restored those cases and added concurrency, two-user token isolation, enumeration, and plaintext-persistence checks. Similar omissions appeared elsewhere: Pool B's static decision table missed a stale client total, while security generation omitted nested item injection, duplicate JSON keys, and a benign apostrophe control. Pool C missed multi-byte boundary behavior, exponent notation, array-valued roles, and duplicate Authorization headers.

Automation introduced a second class of error. AI-generated URL serialization produced double-slash paths in Pools A and C; Pool B exposed a case-sensitive StudentID mismatch and a broken serialized regular expression. Humans treated these as runner defects, discarded their diagnostic results,

corrected the artifacts, reset fixtures, and reran before classifying SUT failures.

The central lesson is that AI is effective for breadth and repeatable structure, but not as an oracle. Specifications, state observability, generated scripts, and execution evidence must be reviewed independently. Keeping unspecified contracts explicit and separating runner failures from product defects made the collaboration defensible.

Word count: 260

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